

IS5314: EBUSINESS SYSTEM INTEGRATION

Effective Term

Semester A 2025/26

Part I Course Overview

Course Title

eBusiness System Integration

Subject Code

IS - Information Systems

Course Number

5314

Academic Unit

Information Systems (IS)

College/School

College of Business (CB)

Course Duration

One Semester

Credit Units

3

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

Nil

Precursors

Nil

Equivalent Courses

Nil

Exclusive Courses

Nil

Part II Course Details

Abstract

This course aims to introduce concepts and practical issues associated with integrating information systems and applications in organizations. Special emphasis is placed on identifying systems integration requirements, selection of appropriate technologies, and analyzing integration solutions.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if app.)	DEC-A1	DEC-A2	DEC-A3
1	Explain the concepts of SI, benefits & challenges of SI, and implications of different types of SI approaches	35			
2	Discover and creatively model internal and external SI requirements of an organization	30	x	x	x
3	Select and justify suitable technologies (middleware) for a given set of SI requirements	15			
4	Critically analyze SI implementations identifying merits and limitations	20			

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Seminars	Students will learn the concepts of systems integration, modelling of integration requirements and architectures, integration solutions, and best practices supplemented with selected case studies. Additional activities are used to reinforce and practice of various integration concepts and techniques using exercises, discussions and short presentations.	1, 2, 3, 4	

2	Project work	Students would have to complete a group project requiring them to perform activities aimed at capturing integration requirements of an organization in business sector. The group project work will be submitted at different phases for review and comments by the instructor/tutors.	1, 2, 3, 4	
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Assessment Tasks / Activities (ATs)

	ATs	CILO No.	Weighting (%)	Remarks ("- " for nil entry)	Allow Use of GenAI?
1	Class participation	2	15	-	Yes
2	Group project work - report	1, 2, 3	25	10%+15%	Yes
3	Group project work - Presentation	1, 2, 3, 4	15	10%+5%	Yes
4	Term paper	3, 4	25	-	No
5	Test	1	20	-	No

Continuous Assessment (%)

100

Minimum Continuous Assessment Passing Requirement (%)

50

Assessment Rubrics (AR)**Assessment Task**

AT1: Class participation (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Ability to collaboratively discover and creatively model internal and external SI requirements of an organization

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT2: Group project work – report (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Ability to explain the concepts of SI, benefits & challenges of SI, and implications of different types of SI approaches

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT2: Group project work – report (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Ability to collaboratively discover and creatively model internal and external SI requirements of an organization

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT2: Group project work – report (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Ability to select and justify suitable technologies (middleware) for a given set of SI requirements

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT3: Group project work – Presentation (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Organization, Use of material; Coverage

- Well-structured and presented in a logical sequence, used time wisely
- The group collectively covered all the major topics/issues.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT3: Group project work – Presentation (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Response to questions and comments

- The group collectively demonstrates full knowledge by answering all questions with explanation and elaboration

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT3: Group project work – Presentation (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Individual - comprehensibility, eye contact and elocution

- Clear and easily understood
- Maintains eye contact with audience, seldom returns to notes
- Uses a clear voice and modulation so that all audience can hear presentation

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT4: Term paper (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Ability to select and justify suitable technologies (middleware) for a given set of SI requirements

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT4: Term paper (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Capability to critically analyze SI implementations identifying merits and limitations

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT5: Test (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Ability to explain the concepts of SI, benefits & challenges of SI, and implications of different types of SI approaches

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT1: Class participation (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Ability to collaboratively discover and creatively model internal and external SI requirements of an organization

Excellent

(A+, A, A-) High

Good

(B+, B) Moderate

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT2: Group project work – report (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Ability to explain the concepts of SI, benefits & challenges of SI, and implications of different types of SI approaches

Excellent

(A+, A, A-) High

Good

(B+, B) Moderate

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT2: Group project work – report (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Ability to collaboratively discover and creatively model internal and external SI requirements of an organization

Excellent

(A+, A, A-) High

Good

(B+, B) Moderate

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT2: Group project work – report (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Ability to select and justify suitable technologies (middleware) for a given set of SI requirements

Excellent

(A+, A, A-) High

Good

(B+, B) Moderate

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT3: Group project work – Presentation (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Organization, Use of material; Coverage

- Well-structured and presented in a logical sequence, used time wisely
- The group collectively covered all the major topics/issues.

Excellent

(A+, A, A-) High

Good

(B+, B) Moderate

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT3: Group project work – Presentation (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Response to questions and comments

- The group collectively demonstrates full knowledge by answering all questions with explanation and elaboration

Excellent

(A+, A, A-) High

Good

(B+, B) Moderate

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT3: Group project work – Presentation (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Individual - comprehensibility, eye contact and elocution

- Clear and easily understood
- Maintains eye contact with audience, seldom returns to notes
- Uses a clear voice and modulation so that all audience can hear presentation

Excellent

(A+, A, A-) High

Good

(B+, B) Moderate

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT4: Term paper (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Ability to select and justify suitable technologies (middleware) for a given set of SI requirements

Excellent

(A+, A, A-) High

Good

(B+, B) Moderate

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT4: Term paper (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Capability to critically analyze SI implementations identifying merits and limitations

Excellent

(A+, A, A-) High

Good

(B+, B) Moderate

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

AT5: Test (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Ability to explain the concepts of SI, benefits & challenges of SI, and implications of different types of SI approaches

Excellent

(A+, A, A-) High

Good

(B+, B) Moderate

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Part III Other Information

Keyword Syllabus

Introduction to eBusiness systems integration

Integration Requirements & Architectures

- Modeling integration requirements
- Enterprise integration architecture
- Technical integration architecture
- Service integration architecture
- Information integration architecture
- Process integration architecture

Integration Solutions

- Application integration solutions and technologies
- Information integration
- Composite application integration
- Process-driven integration

Integration Strategy and Best Practices

- Integration strategies
- Best practices for enterprise integration

Reading List

Compulsory Readings

Title	
1	Nil

Additional Readings

	Title
1	Beth Gold-Bernstein and William Ruth, Enterprise Integration: The Essential Guide to Integration Solutions Addison-Wesley Professional, 2005 (Course textbook)
2	Dirk Krafzig, Karl Banke, Dirk Slama, Enterprise SOA: Service-oriented Architecture Best Practices, Prentice Hall PTR, 2005.
3	Thomas Erl, Service-Oriented Architecture: Concepts, Technology, and Design, Prentice Hall PTR, 2005.
4	Alex Berson and Larry Dubov, Master Data management and Customer Data Integration for a Global Enterprise, McGraw Hill, 2007.
5	Matjaz B. Juric, Benny Mathew, Poornachandra Sarang, Business Process Execution Language for Web Services: BPEL and BPEL4WS, Packt Publishing, 2004.
6	Markus, M. L., Paradigm Shifts - E-Business and Business/Systems Integration, Communications of the AIS, 4:10, November 2000.
7	Hasselbring, W., Information system integration, Communications of the ACM, 43:6, June 2000, pp. 32 - 38.
8	Linthicum, D., Next Generation Application Integration: From Simple Information to Web Services, 2003: Addison-Wesley Professional.
9	Trowbridge, D., et al., Integration Patterns. Patterns & Practices 2004, Microsoft Press.
10	Hohpe, G. and B. Woolf, Enterprise integration patterns, 2004: Addison-Wesley Boston.
11	Umar, A., e-Business and Distributed Systems Handbook (Overview module and Integration module), NGE Solutions, 2003.